



Development and characterization novel bio-adhesive for wood using kenaf core (*Hibiscus cannabinus*) lignin and glyoxal

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ABSTRACT

The recent study focused on lignin-phenol-glyoxal (LPG) as an alternative way to replace toxic formaldehyde used in commercially available wood adhesives. The concern of the uses of carcinogenic formaldehyde in wood adhesive industry has become major problem over human health, environmental and economy issues. In this study, lignin isolated from Kenaf (*Hibiscus cannabinus*) via soda and Kraft pulping were modified into SLPG (soda lignin-phenol-glyoxal) and KLPG (Kraft lignin-phenol-glyoxal) adhesives and were compared to phenol-formaldehyde (PF). Complementary analyses such as Fourier Transform Infrared (FTIR) spectroscopy, ¹H and ¹³C Nuclear Magnetic Resonance (NMR) spectroscopy, thermal stability; Thermogravimetric analysis (TGA) and Differential Scanning Calorimetry (DSC) were utilized to characterize all isolated lignin samples. The physical properties of the resins were further characterized in term of viscosity, gel time and total solid content. It was found that soda lignin comprised higher phenolic —OH content and greater molecular weight compared to Kraft lignin. Various molar ratio of adhesives were applied on plywood and were mechanically tested. The 30% (w/w) SLPG has shown to have higher tensile strength and internal bonding stress at 72.08 MPa and 53.83 N mm^{−2} respectively to that of PF.

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1. Introduction

Phenolic resins are one of many versatile polymers yet invented and currently developed into more applications due to their characteristics such as high temperature stability up to 300 °C, high water and chemical stability. Therefore, phenolic resins are found in a myriad of industrial products such as in the production of carbon foams, molding products, electric laminates, circuit boards, and as coatings and adhesives [1]. At present, the implementation of phenolic resin into adhesives is a breakthrough in the manufacturing products due to their resistance to moisture and high mechanical strength [2,3]. The commercially available wood adhesives that mainly produced using phenolic resins in wood-panel industry are phenol-formaldehyde (PF), urea-formaldehyde (UF) and melamine-urea-formaldehyde (MUF) resins [4]. Additionally, phenol-formaldehyde (PF) resin is the most common type among various adhesives used in wood processing industries such as binder [5], plywood, particleboard, fiberboard, oriented strand board (OSB) [6,7], wood furniture, windows and frames [8]. Both phenol and formaldehyde are derived from petroleum-based industry (non-renewable sources) hence, the scarcity of the hydrocarbon source could affect future cost and the availability of these petroleum based adhesives [9]. Moreover, the toxicity and carcinogenicity of both phenol and

formaldehyde contribute to the downside of implementation has become a great concern to the researchers [10]. Therefore, the substitution of phenol and formaldehyde with bio-based adhesives become a promising alternative way in order to produce formaldehyde free wood adhesives [11].

Several studies have done by using bio-based polymers (such as protein, tannin, lignin and polysaccharides) as suggested polymers for wood adhesives [4,12,13]. Nowadays, in the research areas of adhesives, lignin has become a new raw materials of study in order to substitute phenol in PF formulation, partially or completely, due to their comparable structure [14]. Besides, the potential of lignin as based wood adhesives have been discovered by many researchers [14–16]. Lignin comprises of aromatic and high cross-linked structure, similar to that of PF resin network [9]. In consequence, previous studies on lignin-phenol-glyoxal (LPG) resins to prepare wood adhesives have bloomed and extensive among researchers since glyoxal is an ideal green environmental agent dialdehyde that can replaced formaldehyde in wood adhesives formulation [17]. Supplementary, glyoxal is used to eliminate the formaldehyde from the adhesive formulation. Besides, glyoxal is a simplest aliphatic organic compound that is low toxicity of aldehyde (LD₅₀ rat > 2960 mg/kg; LD₅₀ mouse > 1280 mg/kg) [18], non-volatile [13], easy biodegradability [19], and low-cost substance [17]. In addition, the substitution of glyoxalin adhesives formulation has improved the resulted adhesives in term of physical and mechanical properties of the panels produced [20,21], which means increased the internal

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